



May 2025

Duration: 3 Hours

Max. Marks: 100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- _____ is the practice and precautions taken to protect valuable information from unauthorized access, recording, disclosure or destruction.
A) Network Security B) Database Security
C) Information Security D) Physical Security
- _____ technology is used for analyzing and monitoring traffic in network and information flow.
A) Cloud access security brokers (CASBs) B) Managed detection and response (MDR)
C) Network Security Firewall D) Network traffic analysis (NTA)
- _____ is a combined term which encompasses 3 sub-pillars; information privacy, individual privacy, and communication privacy.
A) Digital Integrity B) Digital privacy
C) Digital secrecy D) Digital protection
- In asymmetric key cryptography, the private key is kept by _____.
A) sender B) receiver
C) sender and receiver D) all the connected devices to the network
- The man-in-the-middle attack can endanger the security of the Diffie-Hellman method if two parties are not _____.
A) Joined B) Authenticated
C) Submit D) Separate
- The shift cipher is sometimes referred to as a _____.
A) Move cipher B) Caesar cipher
C) Cipher D) Cipher text
- In Asymmetric-Key Cryptography, although RSA can be used to encrypt and decrypt actual messages, it is very slow if message is _____.
A) Short B) Long
C) Flat D) Thin
- The size of the block header in bitcoin is _____.
A) 36 bytes B) 36 bits
C) 80 bits D) 80 bytes
- The thing that is mandatory for the IoT gateway to enhance the security of the user is _____.
A) Simple and secure installation B) Data network and storage
C) Software Security D) Protocol abstraction

10. The technology which is pivotal for enabling real-time data analysis in smart home IoT devices is _____
 A) 5G networks B) Dial-up internet connections
 C) Bluetooth technology D) Wi-Fi type 4
11. When troubleshooting connectivity issues in a smart home network, the factor which is crucial to examine is _____
 A) The aesthetic placement of devices B) The strength and stability of the network connection
 C) The color of the LED indicators on devices D) The firmware version of non-IoT appliances
12. In a smart home system, the common cause of delays in executing automated tasks is _____
 A) Insufficient device memory B) Network latency and insufficient processing power
 C) Incorrect device placement D) The use of incompatible device models
13. The primary role of a "Digital Certificate" in securing Smart Grid communication is _____
 A) to authenticate the identity of devices on the network B) to encrypt sensitive data
 C) to monitor network traffic D) to detect malicious activity
14. _____ is the illicit transmission of data from inside an organization or personal system to an external location or recipient.
 A) Database hacking B) Data leakage
 C) Data cracking D) Data revealing
15. A major concern regarding the use of wearable health devices in terms of security is _____
 A) lack of data standardization B) potential for data manipulation
 C) limited battery life D) difficulty in device maintenance
16. The role of Cloud in smart grid architecture of IoT is to _____
 A) Secure data B) collect data
 C) manage data D) store data
17. Asymmetric key cryptography algorithm which relies on the difficulty of factoring large composite numbers is _____
 A) RSA B) ECC
 C) ElGamal D) Diffie-Hellman
18. Name of the attack aims to find two different inputs that produce the same hash value is _____
 A) Brute Force Attack B) Collision Attack
 C) Rainbow Table Attack D) Dictionary Attack
19. _____ is crucial for protecting Cyber-Physical Systems.
 A) Network segmentation B) Increasing CPU clock speed
 C) Adding more RAM D) Using analog data storage
20. _____ is a major challenge in deploying CPS securely.
 A) Large-scale data storage B) Lack of real-time monitoring
 C) Vulnerabilities in communication channels D) High processor speed

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) List and explain the requirements for message authentication.	8	L*2	1	1
	b) Explain briefly about meet-in-the-middle attack?	8	L2	2	1
2.	a) Explain briefly, the direct digital signature approach in cyber security system.	8	L2	1	1
	b) How does hash function works?	8	L2	2	1
3.	a) Classify the cyber security services.	8	L2	2	1

- b) Using polyalphabetic cipher algorithm encrypt the original message "WE ARE DISCOVERED SAVE YOURSELF". Use the key DECEPTIVE for encryption.

8 L3 1 1

Unit – II

4. a) What are the current biggest threats to IoT security? What approach is required to address these threats?
 b) Explain the key security concerns associated with SDN architecture.
5. a) With a neat supportive diagram, explain (i) Software Defined Network (SDN) and (ii) SDN architecture.
 b) Explain the possible Smart home security risks and threats.
6. a) What is Automotive Cyber security? Explain Current Cybersecurity threats in the Automotive Industry.
 b) List any eight best practices to be followed in Automotive Cyber Security to address the threats.

8 L2 3 1
 8 L2 3 1
 8 L2 4 1
 8 L2 4 1
 8 L2 3 1
 8 L2 3 1

Unit – III

7. a) Explain the properties of symmetric key cryptosystems.
 b) Explain with a suitable example, triple data encryption standard (3DES) algorithm used in symmetric key cryptography. Also, indicate its features.
8. a) Explain the implementation issues: cybersecurity, interoperability, scalability, and safety with cyber physical systems (CPS).
 b) What is intelligent cyber physical systems (ICPS)? List and explain the applications of ICPS.

8 L2 5 1
 8 L2 5 1
 8 L2 5 1
 8 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
